

Curriculum					
Modules	Topic	Sub topic	Detail	Outcome	Duration Min
1	Fundamentals	Network Protocols	TCP/IP, HTTP, WebSockets	Understand core network protocols	30
1		Storage	Databases, File Systems	Grasp storage options and trade-offs	30
1		Latency, Throughput, Availability	Definitions, importance	Comprehend key system metrics	30
2	Scalability	Horizontal vs Vertical Scaling	Pros and cons	Understand scaling strategies	30
2		Load Balancing	Algorithms, implementation	Apply load balancing techniques	30
2		Caching	Strategies, cache invalidation	Implement effective caching	30
3	Databases	SQL vs NoSQL	Use cases, trade-offs	Choose appropriate database types	30
3		Replication	Master-slave, multi-master	Implement database replication	30
3		Sharding	Strategies, challenges	Design database sharding solutions	30
4	Distributed Systems	CAP Theorem	Consistency, Availability, Partition Tolerance	Understand distributed system trade-offs	30
4		Consensus Algorithms	Paxos, Raft	Grasp distributed consensus concepts	30
4		Gossip Protocol	Implementation, use cases	Understand decentralized communication	30
5	Message Queues	Pub/Sub Model	Architecture, use cases	Implement asynchronous communication	30
5		Queue vs Stream	Kafka, RabbitMQ	Choose appropriate messaging system	30
6	API Design	REST	Principles, best practices	Design RESTful APIs	30
6		GraphQL	Schema, resolvers	Understand GraphQL API design	30
6		gRPC	Protobuf, use cases	Grasp high-performance RPC	30
7	Microservices	Architecture Patterns	Saga, CQRS	Apply microservices patterns	30
7		Service Discovery	Client-side, server-side	Implement service discovery	30
7		API Gateway	Routing, aggregation	Design API gateway solutions	30
8	Security	Authentication	OAuth, JWT	Implement secure authentication	30
8		Encryption	Symmetric, asymmetric	Apply encryption techniques	30
8		HTTPS, TLS	Handshake process	Secure network communications	30
9	Monitoring	Logging	ELK stack, centralized logging	Implement effective logging	30
9		Metrics	Prometheus, Grafana	Set up system metrics monitoring	30
9		Tracing	Jaeger, Zipkin	Implement distributed tracing	30
10	Design Patterns	Proxy, Facade, Adapter	Use cases, implementation	Apply structural design patterns	30
10		Observer, Strategy, Factory	Use cases, implementation	Apply behavioral design patterns	30
11	Case Studies	URL Shortener	Design considerations	Apply system design principles	30
11		Social Media Platform	Scalability challenges	Design large-scale systems	30